

Pharmaceutical Reform and Pharmacy Entry: Evidence from Hospitals in China

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Empirical Strategy: TWFE

$$Y_{ict} = \beta_0 + \beta_1 \text{Reform}_{ct} + \delta_i + \delta_t + \epsilon_{ict}$$

- ▶ Y_{ict} : Number of pharmacies around hospital i in county c at time t
- ▶ Reform_{ct} : Policy exposure at the county-year level
- ▶ δ_i : Hospital fixed effects
- ▶ δ_t : Year fixed effects

Traditional TWFE may be biased under staggered adoption

Empirical Strategy: CS-DID

$$ATT(g, t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g]$$

- ▶ g : First year of reform (cohort)
- ▶ t : Year
- ▶ Compare treated units with not-yet-treated units
- ▶ Allows for heterogeneous treatment effects
- ▶ Avoids bias from two-way fixed effects under staggered adoption

Estimate dynamic and cohort-specific treatment effects

Baseline DID Results: TWFE (Hospital Cluster)

OLS

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Sample: score=10 | score=8
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors: clustered at hospital level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
did	-0.433*** (0.103)	-0.942*** (0.263)	-1.101* (0.449)
N	193010	193010	193010
R-sq	0.884	0.895	0.903

Baseline DID Results: TWFE (County Cluster)

OLS

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors: clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
did	-0.433** (0.161)	-0.947* (0.441)	-1.113 (0.776)
N	192868	192868	192868
R-sq	0.884	0.895	0.903

Baseline DID Results: CS-DID

Callaway and Sant' Anna

- ▶ Standard errors clustered at the hospital level

Baseline DID Results: PPML

Callaway and Sant' Anna

- ▶ Outcome: Pharmacy entry around hospitals (within 1km)
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors clustered at the hospital level

Event Study: TWFE

TWFE

- ▶ Outcome: Pharmacy entry around hospitals (within 1km)
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors clustered at the hospital level

Event Study: CS-DID

Callaway and Sant' Anna

- ▶ Outcome: Pharmacy entry around hospitals (within 1km)
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors clustered at the hospital level

Summary

- ▶ DID: Positive and statistically significant effect
- ▶ Event Study: No pre-trend, supporting causal inference

The reform increases pharmacy entry around hospitals